Do speech benchmarks measure cognition, or how the speech was transcribed?

1 Two conventions

Convention 1

`Pat: cat (3 seconds)`

Convention 2

`cat dog lion horse cow`

sessions with convention 1

2 Text models learn them

removing annotations

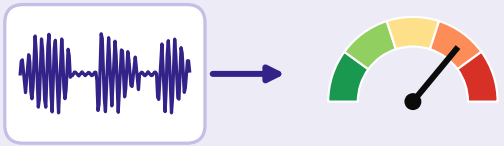
AUC −0.07 to −0.12

timed pauses −0.06 · timer −0.03

only where annotations exist

3 Annotation-free control

ASR + VAD → healthy norms



audio only

risk score

0.81

test AUC [0.70, 0.91]

same under both conventions

Measure pauses from audio · document transcription conventions · report annotation-free controls